INTRODUCTION TO SOFTWARE

LECTURE 5 : WEEK 11

Credit : (02) / Week

T AND REF. BOOKS

Text Book:

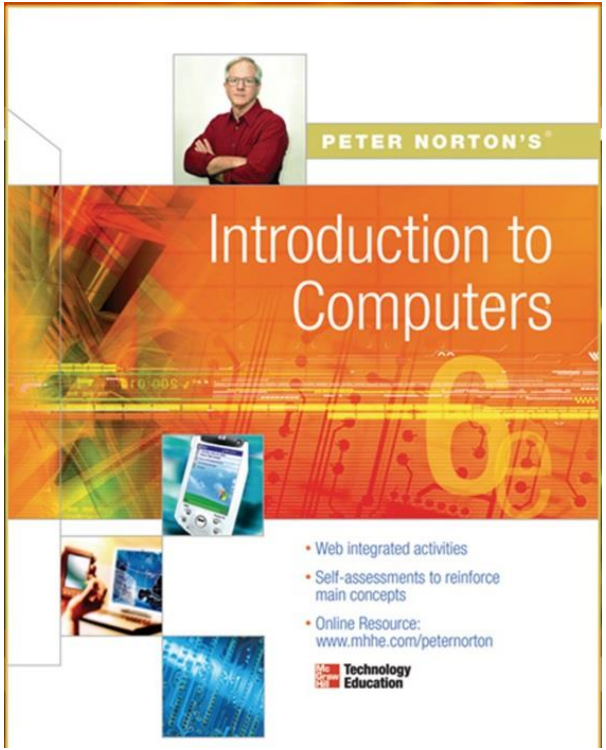
Peter Norton (2011), **Introduction to Computers**, 7 /e, McGraw-Hill

Reference Book:

Gary B (2012), **Discovering Computers**, 1/e, South Western

Deborah (2013), **Understanding Computers**, 14/e, Cengage Learning

June P & Dan O (2014), **New Perspective on Computer**, 16/e



MOBILE ALERT

Kindly **Switch Off** your Mobile/Cell Phone

OR

Switch it to **Silent Mode** Please

WORKING WITH SOFTWARE

SYSTEM SOFTWARE & APPLICATION SOFTWARE

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Learning Outcome

- What is Software
- Software Types
  - System Software
  - Application Software (Word processor, Spreadsheets, Presentation etc)
  - Open Source Software
  - Proprietary Software
  - Utility Software
- Malicious Software
- Acquiring a Software

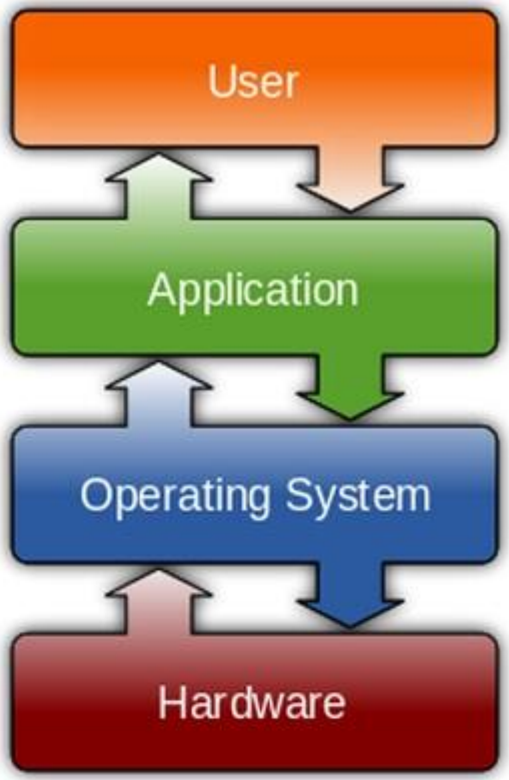
What is a Software ?

- **Computer software**, or simply **software** is any set of machine-readable instructions that directs a computer's processor to perform specific operations.

- Software tells the computer what to do and how to do.

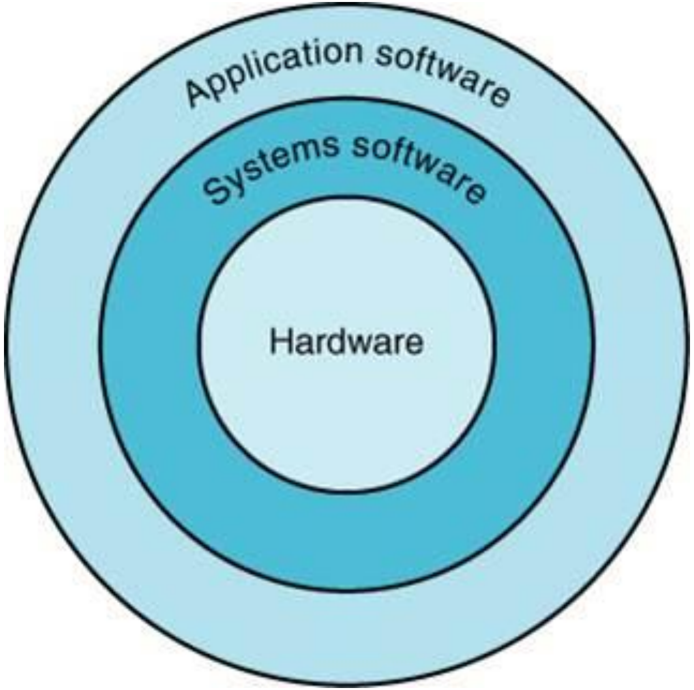
Types of Software

- System Software
- Application Software
- Open Source Software
- Proprietary Software
- Malicious Software



System Software

- A program or set of programs that is especially designed to control different operations of computer system is called system software. It controls the working of different components of the computer.



System Software

- **System software** is computer **software** designed to operate and control the computer hardware and to provide a platform for running **application software**
- **System software** can be separated into two different categories, operating **systems** and utility **software**

System Software

- System Software includes the Operating System (OS) and all the utilities that enable the computer to function.
- System software is a term referring to any computer software which manages and controls the hardware so that application software can perform a task.

Example:

Operating Systems, Device Drivers, Interpreters

Application Software

- It is a set of one or more programs designed to carry out **operations** for a **specific application**. Application software **cannot run on itself** but is dependent on system software to execute.

- Example:

  Payroll systems, Inventory Control,

  Word Processor, Spreadsheet

  and Database Management System etc.,

Open Source Software

- Open source software (OSS) is computer software whose **source code** is available under a **license** that permits users to **use, change, and improve** the software, and to redistribute it in modified or unmodified form.

- It is often developed in a public, collaborative manner.
  - Well-known OSS products are Linux, Netscape, Apache, etc.,

Open Source Software

Proprietary Software

- **Proprietary software (non-free software)**
  - Software with **restrictions** on **using**, **copying** and **modifying** as enforced by the proprietor. Restrictions on use, modification and copying is achieved by either legal or technical means and sometimes both.
- Proponents (advocate) of proprietary software are Microsoft.
- Example : MS Office/Windows, CAD, Norton AV etc.,

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Types of System Software

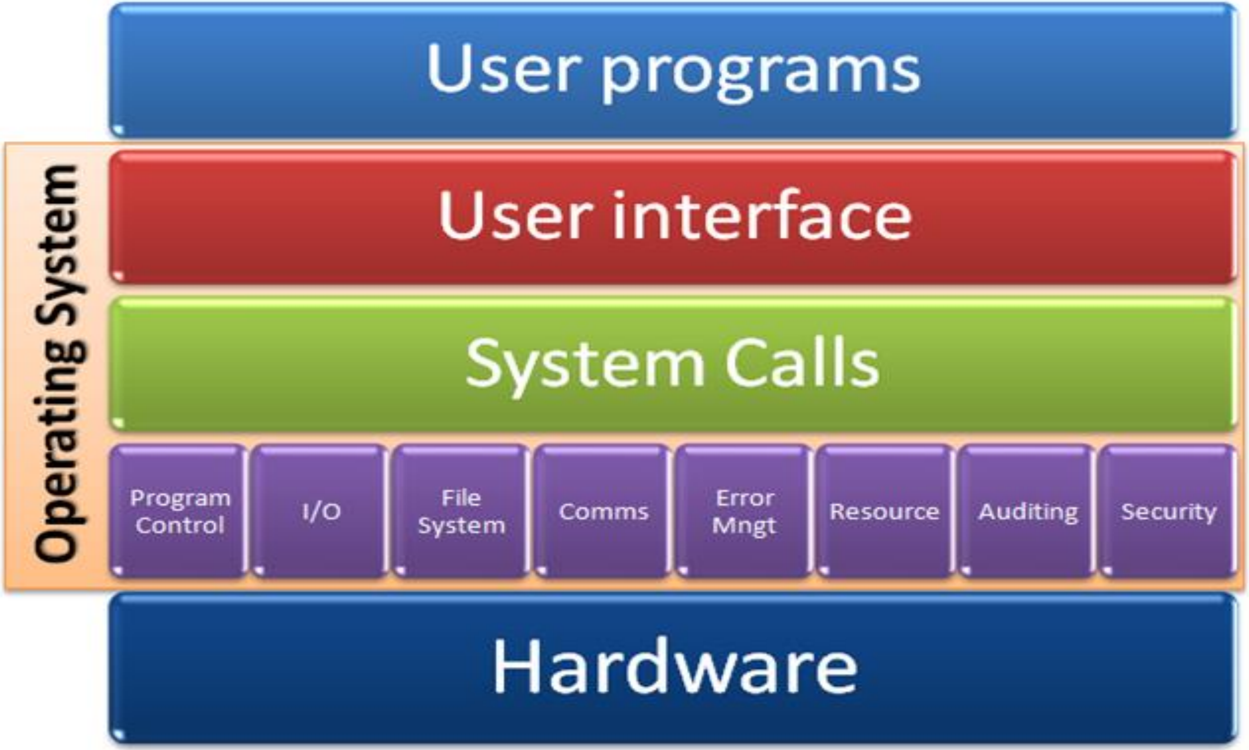
- OS Software
- Utilities
- Device driver

Operating System

- An operating system **(OS)** is software that **manages** computer **hardware and software** resources and provides **common services** for computer programs.
- Various characteristics of OS are described as

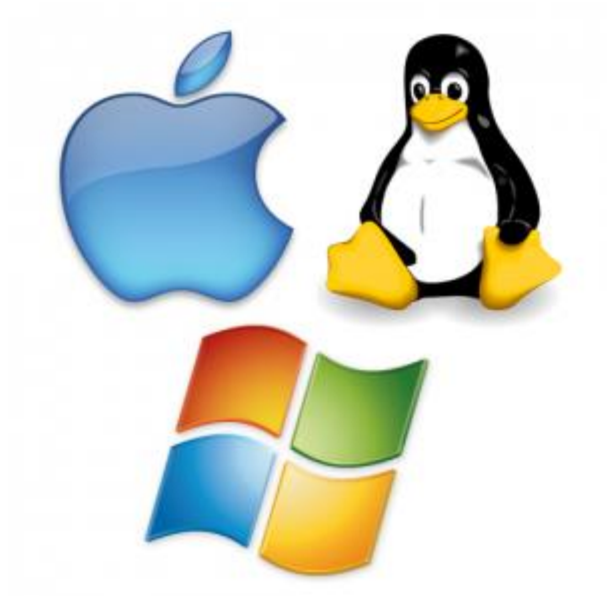
Operating System

- Operating systems perform basic tasks, such as recognizing input from the keyboard, sending output to the display screen, keeping track of files and directories on the disk, and controlling peripheral devices such as disk drives and printers


User programs
Operating System
User interface
System Calls
Program Control
I/O
File System
Comms
Error Mngt
Resource
Auditing
Security
Hardware

Functions of Operating Systems

- Provide a user interface
- Run programs
- Manage hardware devices
- Organized file storage
- Memory management



Characteristics of Operating systems

- **Multi-user**

  Allows two or more users to run programs at the same time. Some operating systems permit **hundreds** or even **thousands** of concurrent users

- **Multi-tasking**

  Allows a user to perform more than one computer **task** at a time.

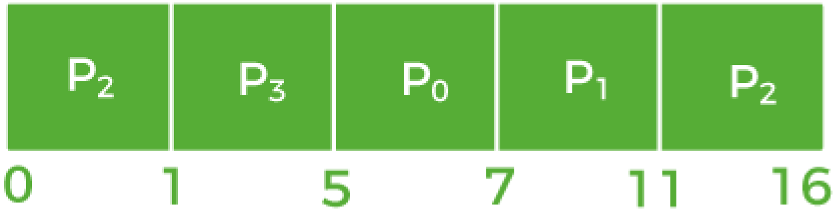
MULTI TASKING

- **There are two types of multitasking:**
- Preemptive multitasking. In preemptive multitasking, the operating system decides how to allocate CPU time slices to each program. ...
- Cooperative multitasking. In cooperative multitasking, each program controls how much CPU time it needs.

PREEMTIVE MULTI TASKING

Let us take the example of Preemptive Scheduling. We have taken P0, P1, P2, and P3 are the four processes.

Process	Arrival Time	CPU Burst time (in millisec.)
P0	3	2
P1	2	4
P2	0	6
P3	1	4



Preemptive Scheduling

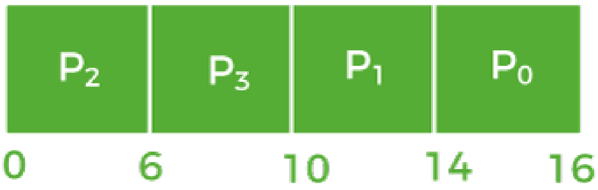
PRE EMTIVE MULTI TASKING

- Firstly, the process **P2** comes at time **0**. So, the CPU is assigned to process **P2**.
- When process **P2** was running, process P3 arrived at time 1, and the remaining time for process **P2 (5 ms)** is greater than the time needed by process **P3 (4 ms)**. So, the processor is assigned to P3.
- When process **P3** was running, process **P1** came at time **2**, and the remaining time for process **P3 (3 ms)** is less than the time needed by processes **P1 (4 ms)** and **P2 (5 ms)**. As a result, **P3** continues the execution.
- When process **P3** continues the process, process **P0** arrives at time **3**. P3's remaining time **(2 ms)** is equal to **P0's** necessary time **(2 ms)**. So, process **P3** continues the execution.
- When process **P3** finishes, the CPU is assigned to **P0**, which has a shorter burst time than the other processes.
- After process **P0** completes, the CPU is assigned to process **P1** and then to process **P2**.

NON PRE EMTIVE MULTI TASKING

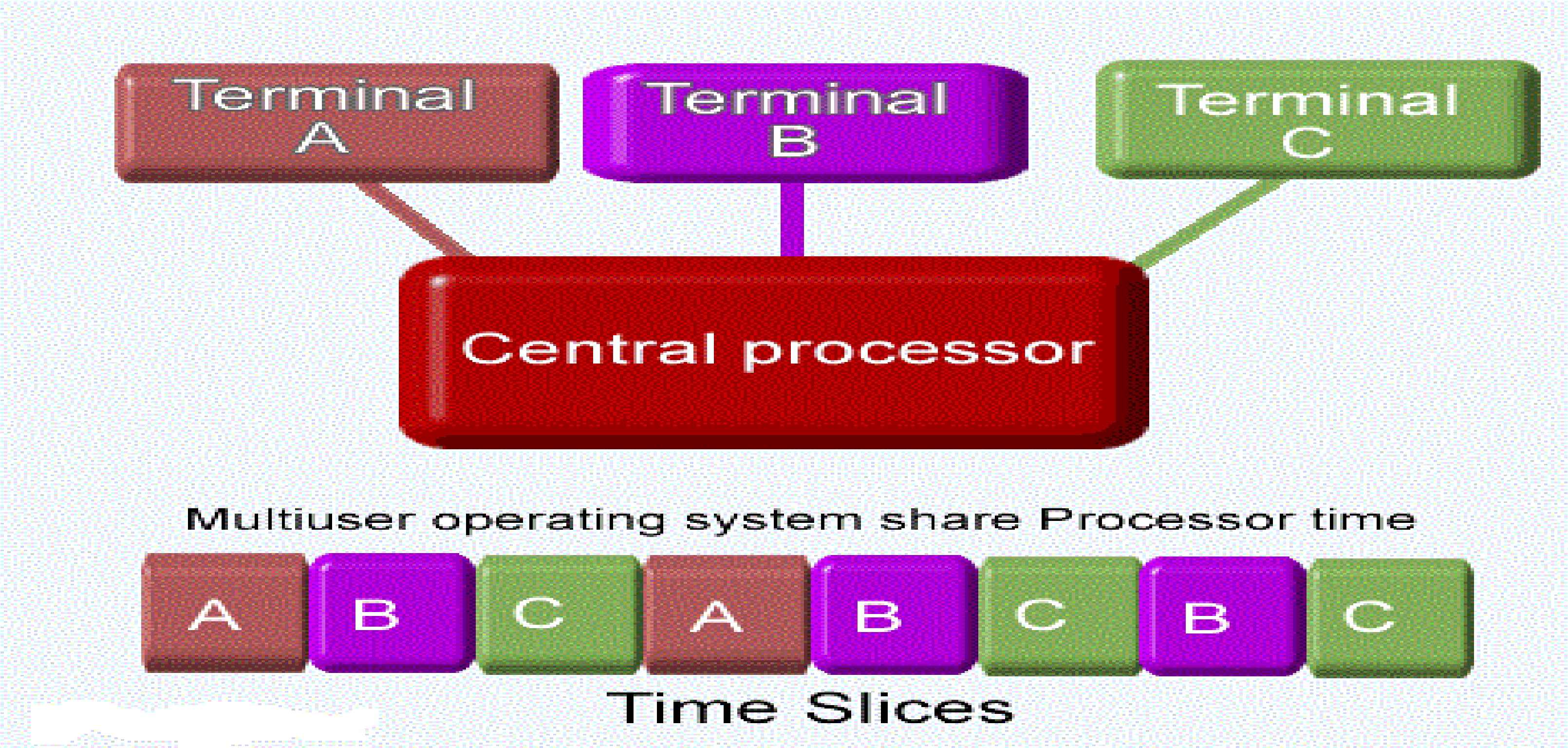
Let's take the above preemptive scheduling example and solve it in a non-preemptive manner.

•The process **P2** comes at **0**, so the processor is assigned to process **P2** and takes **(6 ms)** to execute.
•All of the processes, **P0, P1**, and **P3**, arrive in the ready queue in between. But all processes wait till
•process **P2** finishes its CPU burst time.
•After that, the process that comes after process **P2**, i.e., **P3**, is assigned to the CPU until it finishes its burst time.
•When process **P1** completes its execution, the CPU is given to process **P0**.



Non - Preemptive Scheduling

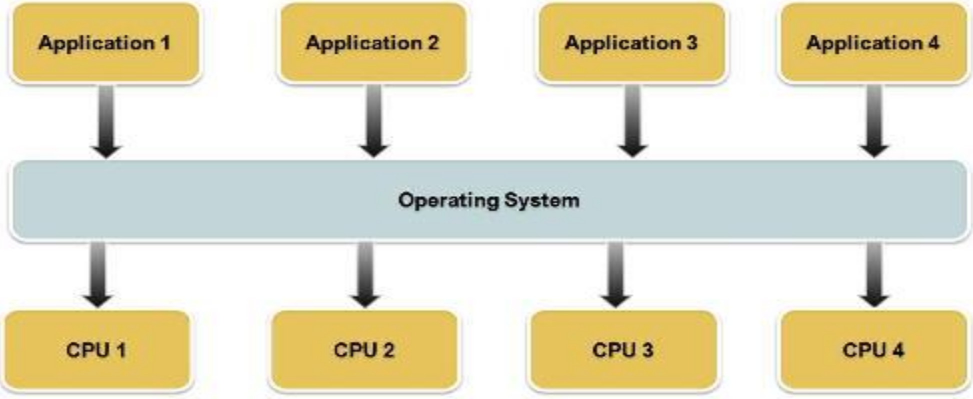
Multi-user / Multi tasking



Characteristics of Operating systems

- **Multiprocessing**

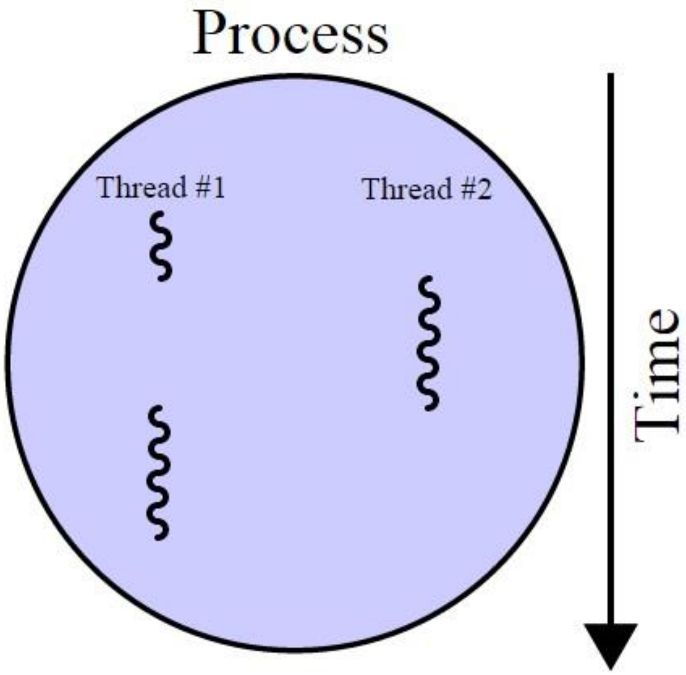
  Use of two or more **central processing units** (CPUs) within a single computer system for running a program or application



Characteristics of Operating systems

- Multithreading

  The ability of an operating system to execute different parts of a program, called ***threads***, simultaneously



Characteristics of Operating systems

- **Real time**

  A system is said to be **Real Time** if it is required to complete it's work & deliver it's **services on time** or instantly

  - Example – Flight Control System
  - All tasks in that system must execute on time.
    - Very fast small OS
    - Built into a device
    - Respond quickly to user input

Characteristics of Operating systems

- **Hard Real Time System**
  - Failure to meet deadlines is fatal
  - example : Flight Control System

- **Soft Real Time System**
  - Late completion of jobs is undesirable but not fatal
  - System performance degrades as more & more jobs miss deadlines
  - Online Databases

Types of Operating Systems

- Real-time operating system

A system is said to be **Real Time** if it is required to complete it's work & deliver it's **services on time** or instantly

- Example – Flight Control System, Medical devices
- All tasks in that system must execute on time
  - Very fast small OS
  - Built into a device
  - Respond quickly to user input

Types of Operating Systems

- Single user/Single tasking OS
  - One user works on the system
  - Performs one task at a time
  - MS-DOS and Palm OS
  - Take up little space on disk
  - Run on inexpensive computers

Types of Operating Systems

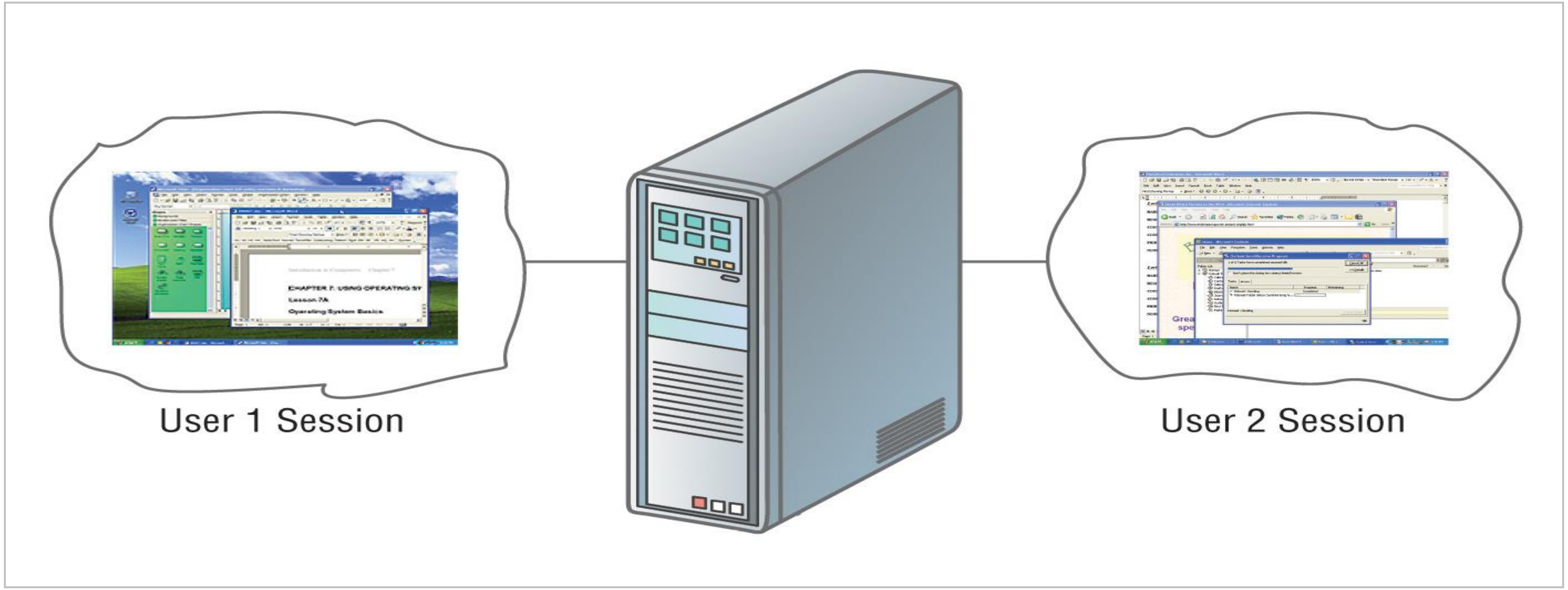
- Single user/Multitasking OS
  - User performs many tasks at once
  - Most common form of OS
  - Windows XP/7/8 and OS X
  - Require expensive computers
  - Tend to be complex
- Multi-tasking operating systems are complex
- However both XP and OS X make the multitasking process painless

Types of Operating Systems

- Multi user/Multitasking OS

  Found on supercomputers, mainframes and minicomputers
  - Many users connect to one computer
  - Each user has a unique session
  - UNIX, Linux, and VMS
  - Maintenance can be easy
  - Requires a powerful computer

Multi user/Multi tasking OS



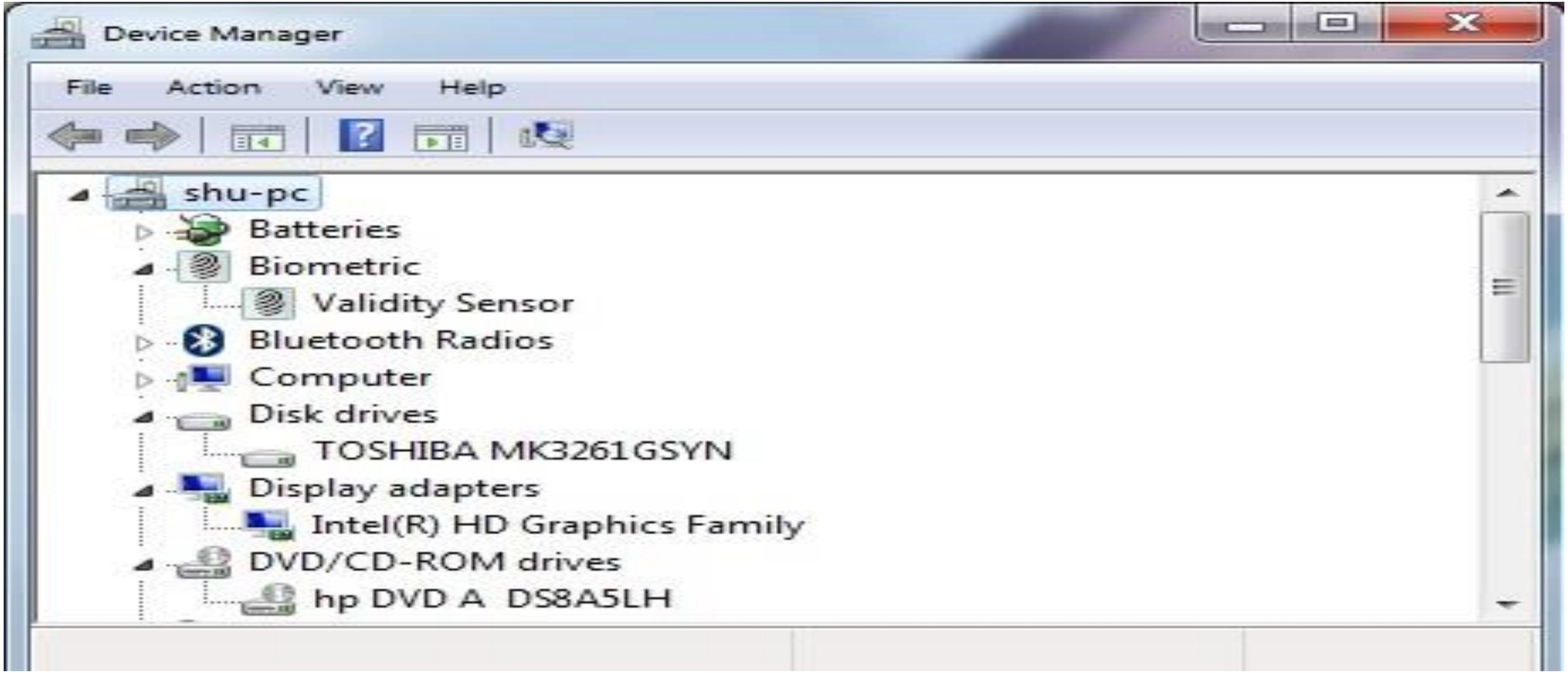
Classification of Operating systems

- **Multi-user:** Allows two or more users to run programs at the same time. Some operating systems permit hundreds or even thousands of concurrent users.
- **Multiprocessing :** Supports running a program on more than one CPU.
- **Multitasking :** Allows more than one program to run concurrently.
- **Multithreading :** Allows different parts of a single program to run concurrently.
- **Real time:** Responds to input instantly.

Device Driver

- A **device driver** (commonly referred to as simply a **driver**) is a computer program that operates or controls a particular type of device that is attached to a computer.
- Examples Mouse driver, keyboard driver, video card driver, sound card driver

Device Driver



THANKYOU

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